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LIFE PREDICTION METHOD, LIFE PREDICTION APPARATUS, AND POWER GENERATION SYSTEM

Abstract

A life prediction method according to the present disclosure is a life prediction method for a field cell apparatus including power generation units each including a fuel cell. The life prediction method includes predicting a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period.

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Background/Summary

BACKGROUND

1. Technical Field

[0001] The present disclosure relates to a life prediction method, a life prediction apparatus, and a power generation system.

2. Description of the Related Art

[0002] Various proposals have hitherto been made with respect to the life prediction for power generation systems. In one example, Japanese Unexamined Patent Application Publication NO. 11-97049 discloses a life prediction method for a fuel cell. The life prediction method includes: measuring a rate of change in voltage of a fuel cell operated in accordance with a basic operation pattern (step S1); approximating, by a predetermined approximate expression, the relationship between the rate of change in voltage and the power generation time with respect to the basic operation pattern (step S2); by using the approximate expression, determining the amount of decrease in voltage of the fuel cell (step S3); and in response to the amount of decrease in voltage exceeding a predetermined value, determining that the life of the fuel cell has ended (step S4).

SUMMARY

[0003] One non-limiting and exemplary embodiment provides a life prediction method, a life prediction apparatus, and a power generation system that make it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life.

[0004] In one general aspect, the techniques disclosed here feature a life prediction method for a fuel cell apparatus including power generation units. The power generation units each include a fuel cell. The life prediction method includes predicting a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period.

[0005] The life prediction method, the life prediction apparatus, and the power generation system according to an aspect of the present disclosure make it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life.

[0006] It should be noted that general or specific embodiments may be implemented as a system, a method, an integrated circuit, a computer program, a storage medium, or any selective combination thereof.

[0007] Additional benefits and advantages of the disclosed embodiments will become apparent from the specification and drawings. The benefits and/or advantages may be individually obtained by the various embodiments and features of the specification and drawings, which need not all be provided in order to obtain one or more of such benefits and/or advantages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates an example of a power generation system according to an embodiment;
[0009] FIG. 2 illustrates an example of a life prediction apparatus illustrated in FIG. 1;
[0010] FIG. 3 is a flowchart according to the embodiment, illustrating an example of an operation (method for predicting the life of each of power generation units of a fuel cell apparatus) of the life prediction apparatus in the power generation system;
[0011] FIG. 4 is a flowchart according to Example 1 of the embodiment, illustrating an example of an operation (method for predicting the life of each of power generation units of the fuel cell apparatus) of the life prediction apparatus in the power generation system;
[0012] FIG. 5 is a flowchart according to Example 2 of the embodiment, illustrating an example of an operation (method for predicting the life of each of power generation units of the fuel cell apparatus) of the life prediction apparatus in the power generation system;
[0013] FIG. 6 illustrates an example of a voltage stabilization timing when voltage becomes stable after the elapse of a predetermined time from the start of power generation by a power generation unit;
[0014] FIG. 7 illustrates, for explanatory purposes, an example of an approximation line representing the correlation between the cumulative power generation time of a power generation unit and its voltage; and
[0015] FIG. 8 illustrates an example of a power generation plan for the power generation system according to Example 3 of the embodiment, for explaining an example of a method for determining a ratio R1 of the number of power generation units that generate power out of the total number of power generation units in a predetermined period.

DETAILED DESCRIPTIONS

[0016] Japanese Unexamined Patent Application Publication NO. 11-97049 does not consider predicting the time when multiple power generation units each reach the end of life. Specifically, in Japanese Unexamined Patent Application Publication NO. 11-97049, the operating ratio of a fuel cell apparatus including multiple power generation units is not taken into account. This makes it difficult to accurately predict the time when each of the power generation units reaches the end of life.

[0017] Accordingly, a first aspect of the present disclosure provides a life prediction method for a fuel cell apparatus including power generation units. The power generation units each include a fuel cell. The life prediction method includes predicting a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period.

[0018] The life prediction method according to the first aspect mentioned above makes it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life.

[0019] Specifically, the ratio of the number of power generation units that generate power out of the total number of power generation units of the fuel cell apparatus in the predetermined period corresponds to the number-based operating ratio of the fuel cell apparatus. In other words, the number-based operating ratio of the fuel cell apparatus corresponds to the ratio of the number of power generation units that generate power out of the total number of power generation units when the fuel cell apparatus generates power in the predetermined period.

[0020] In this regard, if each of the power generation units is used substantially evenly in the predetermined period, the number-based operating ratio of the fuel cell apparatus and the capacity utilization ratio of each of the power generation units are substantially equal. The higher the

capacity utilization ratio of a power generation unit, the faster the rate of degradation of the power generation unit. That is, a negative correlation exists between the ratio mentioned above and the life of the power generation unit.

[0021] As described above, the life prediction method according to the first aspect takes the ratio mentioned above into account in predicting the time when each of the power generation units that includes a fuel cell reaches the end of life. This allows for accurate prediction of the time when each of the power generation units reaches the end of life, in comparison to a case where the ratio mentioned above is not taken into account.

[0022] According to a second aspect of the present disclosure, in the life prediction method according to the first aspect, the cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life may be calculated based on a prediction of a decrease in voltage due to degradation of the power generation unit of the fuel cell apparatus.

[0023] A power generation unit is generally known to decrease in voltage due to degradation of the power generation unit as the cumulative power generation time of the power generation unit increases.

[0024] Accordingly, with the life prediction method according to the second aspect, for example, it is predicted, from the progression of a voltage (actual measurement data) with respect to the cumulative power generation time of each power generation unit, how the value of the voltage will decrease with the passage of the cumulative power generation time in the future. Then, the cumulative power generation time at which the predicted value of the voltage reaches a predetermined end-of-life threshold voltage can be calculated appropriately as the cumulative power generation time at which the power generation unit is determined to reach the end of life.

[0025] According to a third aspect of the present disclosure, in the life prediction method according to the first aspect or the second aspect, the predetermined time may include time slots, and the ratio may be calculated as a first value divided by a second value. The first value is a summation, over the predetermined period, of a number of the power generation units that generate power in each of the time slots. The second value is the total number of the power generation units multiplied by a total number of the time slots included in the predetermined period.

[0026] With the life prediction method according to the third aspect mentioned above, the number-based operating ratio of the fuel cell apparatus in the predetermined period can be calculated from the following values: the summation of the number of power generation units that are “required to generate power” in each of the time slots; and the summation of the total number of power generation units in each of the time slots.

[0027] A fourth aspect of the present disclosure provides a life prediction apparatus for a fuel cell apparatus including power generation units. The power generation units each include a fuel cell. The life prediction apparatus includes a controller, and a communicator. The controller predicts a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period. The communicator communicates information to a display. The information indicates the predicted time when the power generation unit of the fuel cell apparatus reaches the end of life.

[0028] The life prediction apparatus according to the fourth aspect configured as described above makes it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life. Details of an operational effect provided by the life prediction apparatus according to the fourth aspect will not be described below as the operational effect is substantially the same as the operational effect provided by the life prediction method according to the first aspect.

[0029] A fifth aspect of the present disclosure provides a power generation system including a fuel cell apparatus, and the life prediction apparatus according to the fourth aspect. The fuel cell apparatus includes power generation units. The power generation units each include a fuel cell.

[0030] The configuration mentioned above makes it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life. Details of an operational effect provided by the power generation system according to the fifth aspect will not be described below as the operational effect is substantially the same as the operational effect provided by the life prediction method according to the first aspect.

[0031] Specific examples of the above-mentioned aspects of the present disclosure will now be described with reference to the attached drawings. The specific examples described below each represent one example of the above-mentioned aspects of the present disclosure. Specific details set forth in the following description, such as shapes, numeric values, constituent elements, and the positioning and connection of constituent elements, are not intended to limit the scope of the claims unless specifically recited in the claims.

[0032] Those constituent elements in the following description which are not cited in the independent claim representing the most generic concept of the present disclosure will be described as optional constituent elements. In the drawings, descriptions of features designated by the same reference signs will be sometimes omitted. For ease of understanding, the drawings schematically depict individual constituent elements, and are sometimes not accurate representations as to their shapes, dimensional ratios, or other details.

[0033] Further, as for apparatus operation, as required, the order of steps may be rearranged, or a known step may be added.

EMBODIMENT

Apparatus Configuration

[0034] FIG. 1 illustrates an example of a power generation system according to an embodiment. FIG. 2 illustrates an example of a life prediction apparatus illustrated in FIG. 1.

[0035] A power generation system **10** according to the embodiment includes a fuel cell apparatus **15**, and a life prediction apparatus **20**. The fuel cell apparatus **15** includes power generation units each including a fuel cell. As illustrated in FIG. 1, the power generation system **10** includes power generation units grouped into the following groups of power generation units each including a fuel cell: power generation units a1 to an, b1 to bn, c1 to cn, d1 to dn, and e1 to en. The power generation system **10** may be, for example a system that supplies bulk power to a power system. The configuration of the fuel cell apparatus **15** in FIG. 1 will now be described in more detail.

[0036] The power generation units are grouped into the power generation units a1 to an, b1 to bn, c1 to cn, d1 to dn, and e1 to en. Although not illustrated, the power generation units a1 to an, b1 to bn, c1 to cn, d1 to dn, and e1 to en each include components such as a fuel cell stack, a power conditioner, and a control device. The power conditioner serves to convert direct-current power generated by the fuel cell stack into alternating-current power for output to the power system. The control device controls operation of the fuel cell stack and the power conditioner.

[0037] In the present example, the power generation units are grouped into the following groups: the power generation units a1 to an belonging to a fuel cell apparatus **15A**; the power generation units b1 to bn belonging to a fuel cell apparatus **15B**; the power generation units c1 to cn belonging to a fuel cell apparatus **15C**; the power generation units d1 to dn belonging to a fuel cell apparatus **15D**; and the power generation units e1 to en belonging to a fuel cell apparatus **15E**. All power generation units belonging to one group are also referred to simply as “power generation units within a group.” Hereinafter, for convenience of explanation, the power generation units a1 to an, b1 to bn, c1 to cn, d1 to dn, and e1 to en will be sometimes abbreviated as “power generation units **15ij** (i=a to e, j=1 to n).”

[0038] The configuration of the fuel cell apparatus **15** mentioned above is, however, intended to be

illustrative and not limiting. For example, the fuel cell apparatus **15** may be grouped based on power generation units within a single group.

[0039] Control devices **30A** to **30E** are provided respectively for a group A, a group B, a group C, a group D, and a group E, in other words, for the fuel cell apparatuses **15A** to **15E**. The control devices **30A** to **30E** each control operation of each individual power generation unit within the corresponding group.

[0040] For example, the control device **30A** controls, via a communication network, the output of each of the power generation units **a1** to **an** belonging to the group A so as to allow efficient operation (e.g., life optimization) of the power generation units **a1** to **an**. In one configuration, each power generation unit may include no control device, and the operation of power generation units belonging to individual groups may be controlled directly by the control devices **30A** to **30E**.

[0041] The control devices **30A** to **30E** may be any devices with a control capability. The control devices **30A** to **30E** each include a processing section (not illustrated), a storage section (not illustrated) that stores a control program, and a communicator (not illustrated). As the processing section reads and executes the control program stored in the storage section, a predetermined control is executed in each of the control devices **30A** to **30E**. An example of the processing section is a microprocessor. An example of the storage section is a memory.

[0042] The life prediction apparatus **20** predicts the life of each individual power generation unit included in the fuel cell apparatus **15**. As illustrated in FIG. 2, the life prediction apparatus **20** includes a communicator **21**, and a controller **23**.

[0043] The controller **23** predicts the time when each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) reaches the end of life (hereinafter, “end-of-life time”) from the following values: the cumulative power generation time at which each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) is determined to reach the end of life; and a ratio **R1** of the number of power generation units that generate power out of the total number of power generation units in a predetermined period.

[0044] The meaning of the above-mentioned expression “the number of power generation units that generate power out of the total number of power generation units” may include not only power generation units that are determined as being “required to generate power” in a power generation plan stored in a storage section of the controller **23**, but also power generation units that have actually generated power.

[0045] The “predetermined period” mentioned above may be the entire period of the power generation plan stored in the storage section of the controller **23**, or may be part of the entire period. Although the term “predetermined period” may mean, for example, a period of about one to several days, the term may also mean a suitable period with a longer duration ranging from one week (seven days) to one month.

[0046] As for the “cumulative power generation time at which each of the power generation units is determined to reach the end of life” and the “ratio **R1** of the number of power generation units that generate power out of the total number of power generation units in a predetermined period”, their respective specific examples will be described later in the description of Examples.

[0047] The communicator **21** is a transmitter that communicates, to a display **40**, information indicating the predicted end-of-life time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15**. The information may be, for example, a year/month/day on a calendar displayed on the display **40**. Examples of the “display **40**” may include, but are not limited to, an information terminal of a service consumer who receives the supply of power generated by the power generation system **10**, and a display device of a maintenance company. Accordingly, for example, a service consumer who receives the supply of power generated by the power generation system **10** can easily learn how much life each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) has.

[0048] In the present example, the controller **23** predicts the end-of-life time for each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n). This, however, is not intended to be limiting.

[0049] In another alternative example, control devices (e.g., the control devices **30A** to **30E**) other than the controller **23** may predict the end-of-life time for each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n).

[0050] In another alternative example, control devices (e.g., control devices (not illustrated) within individual power generation units) other than the controller **23** and the control devices **30A** to **30E** may predict the end-of-life time for each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n).

[0051] The controller **23** may be any component with a control capability. The controller **23** includes a processing section (not illustrated), and a storage section (not illustrated) that stores a control program. As the processing section reads and executes the control program stored in the storage section, a predetermined control is executed in the controller **23**. An example of the processing section is a microprocessor. An example of the storage section is a memory.

[0052] The configuration of the power generation system **10** mentioned above is, however, intended to be illustrative and not limiting. For example, the life prediction apparatus **20** may be integrated with the control devices **30A** to **30E**, in other words, the life prediction apparatus **20** may incorporate the control capability of the control devices **30A** to **30E** to directly control the operation of each of power generation units within the corresponding groups.

Operation

[0053] FIG. **3** is a flowchart according to the embodiment, illustrating an example of an operation (method for predicting the life of each power generation unit of the fuel cell apparatus) of the life prediction apparatus in the power generation system. The operation described below may be performed by, for example, the processing section of the controller **23** reading the control program from the storage section of the controller **23**. However, the operation described below does not necessarily need to be performed by the controller **23**. An operator may perform part of the operation. The example described below is directed to a case where the operation is controlled by the controller **23**.

[0054] When operation of the life prediction apparatus **20** starts, then at step **S1**, the end-of-life time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15** is predicted from the following values: the cumulative power generation time at which each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) is determined to reach the end of life; and the ratio **R1** of the number of power generation units that generate power out of the total number of power generation units in a predetermined period.

[0055] Specifically, based on the ratio **R1**, it is estimated how much the cumulative power generation time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15** will increase relative to how much the cumulative power generation time of the fuel cell apparatus **15** will increase in the future, and then it is predicted when the cumulative power generation time determined to be the end of life is reached.

[0056] In the above prediction, a time-based operating ratio **R2**, which represents the ratio of the operating time of the fuel cell apparatus **15** in a predetermined period, may be taken into account. The time-based operating ratio **R2** will be described later in detail with reference to Example 1.

[0057] For each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n), the operation at step **S1** may be repeated every time the cumulative power generation time increases by a predetermined amount. Information indicating the predicted end-of-life time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) may be communicated to the display **40** (see FIG. **2**). The information may be displayed as, for example, a year/month/day on a calendar on the display **40**.

[0058] The embodiment as described in the foregoing makes it possible to predict, with higher accuracy than in the related art, the time when each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) included in the corresponding fuel cell apparatuses **15A** to **15E** reaches the end of life.

[0059] Specifically, the ratio **R1**, which represents the ratio of the number of power generation units that generate power out of the total number of power generation units of the fuel cell apparatus **15** in the predetermined period, corresponds to the number-based operating ratio of the

fuel cell apparatus **15**. In this regard, if each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) is used substantially evenly in the predetermined period, the number-based operating ratio of the fuel cell apparatus **15** and the capacity utilization ratio of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) are substantially equal. The higher the capacity utilization ratio of a power generation unit, the faster the rate of degradation of the power generation unit. That is, a negative correlation exists between the ratio $R1$ and the life of the power generation unit.

[0060] As described above, according to the embodiment, the ratio $R1$ is taken into account in predicting the time when each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) that includes a fuel cell reaches the end of life. This allows for accurate prediction of the time when each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) reaches the end of life, in comparison to a case where the ratio $R1$ is not taken into account.

Example 1

[0061] FIG. **4** is a flowchart according to Example 1 of the embodiment, illustrating an example of an operation (method for predicting the life of each of power generation units of the fuel cell apparatus) of the life prediction apparatus in the power generation system. The operation described below may be performed by, for example, the processing section of the controller **23** reading the control program from the storage section of the controller **23**. However, the operation described below does not necessarily need to be performed by the controller **23**. An operator may perform part of the operation. The example described below is directed to a case where the operation is controlled by the controller **23**.

[0062] When operation of the life prediction apparatus **20** starts, then at step **S11**, a ratio $R1i$ ($i=a$ to e) is derived for each of the fuel cell apparatuses **15A** to **15E**. The ratio $R1i$ is the ratio of the number of power generation units that generate power out of the total number of power generation units in a predetermined period.

[0063] Subsequently, from the ratio $R1i$ ($i=a$ to e) derived at step **S11**, it is estimated how much the cumulative power generation time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) will increase relative to how much the cumulative power generation time of the fuel cell apparatus **15** will increase in the future. Specifically, the future cumulative power generation time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) is estimated as “the future cumulative power generation time Tij ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15** × the ratio $R1$.” This is explained below with respect to, for example, the ratio $R1a$ for the fuel cell apparatus **15A**. If the ratio $R1a$ is 0.5, then the future cumulative power generation time of each of the power generation units **15aj** ($j=1$ to n) is estimated to be 0.5 times the future cumulative power generation time Ta of the fuel cell apparatus **15A**.

[0064] Then, at step **S12**, the following value is calculated: the future cumulative power generation time Tij ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15** at which the current cumulative power generation time Tij ($i=a$ to e , $j=1$ to n) of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) plus “the future cumulative power generation time Tij ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15** × the ratio $R1i$ ($i=a$ to e)” becomes equal to the cumulative power generation time (end-of-life threshold time TE) at which the end of life is determined to be reached. From the calculated value of the future cumulative power generation time Tij ($i=a$ to e , $j=1$ to n), the end-of-life time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) is predicted.

[0065] For each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n), the operations at steps **S11** and **S12** may be repeated every time the cumulative power generation time increases by a predetermined amount. Information indicating the predicted end-of-life time of each of the power generation units **15ij** ($i=a$ to e , $j=1$ to n) may be communicated to the display **40** (see FIG. **2**). The information may be displayed as, for example, a year/month/day on a calendar on the display **40**.

[0066] The life prediction method for the fuel cell apparatus **15** described in the foregoing is, however, intended to be illustrative and not limiting.

[0067] For example, assuming that in a predetermined period, the fuel cell apparatus **15** is

continuously performing power generation, and expected to continue to perform power generation in a similar manner with no interruption of power generation, then the future cumulative power generation time T_{ij} ($i=a$ to e , $j=1$ to n) calculated in the above-mentioned example, that is, the future cumulative power generation time of the fuel cell apparatus **15** at which the end-of-life threshold time T_E is reached, is used directly in predicting the end-of-life time of each of the power generation units 15_{ij} ($i=a$ to e , $j=1$ to n).

[0068] However, if the predetermined period includes a period during which the fuel cell apparatus **15** stops power generation, a time-based operating ratio R_{2i} ($i=a$ to e) of the fuel cell apparatus **15** may be taken into account in predicting the life of each of the power generation units 15_{ij} ($i=a$ to e , $j=1$ to n). The “time-based operating ratio R_{2i} ($i=a$ to e)” corresponds to the ratio of the time for which each of the fuel cell apparatuses **15A** to **15E** performs power generation (operates) in the predetermined period. In other words, the time-based operating ratio R_{2i} ($i=a$ to e) corresponds to the ratio of the time for which at least one of power generation units included in each of the fuel cell apparatuses **15A** to **15E** performs power generation in the predetermined period.

[0069] This is explained below with respect to, for example, the time-based operating ratio R_{2a} of the fuel cell apparatus **15A**. If a predetermined period in a power generation plan stored in the storage section of the controller **23** is one month, then the time-based operating ratio R_{2a} is calculated based on the following pieces of data: how many hours per day the fuel cell apparatus **15A** performs power generation; and how many days per month the fuel cell apparatus **15A** performs power generation. In one example, if the fuel cell apparatus **15A** performs power generation for only a half-day per day, and does not perform power generation on weekends, then the time-based operating ratio R_{2a} of the fuel cell apparatus **15A** is calculated to be about 0.37 ($=0.5 \times 22/30$).

[0070] Subsequently, the end-of-life time of each of the power generation units 15_{aj} ($j=1$ to n) is predicted by calculating the following value: “the future cumulative power generation time T_{aj} ($j=1$ to n) of the fuel cell apparatus **15** at which the end-of-life threshold time T_E is reached/ R_{2a} .” Specifically, the end-of-life time of each of the power generation units 15_{aj} ($j=1$ to n) is predicted from a value obtained by multiplying, by a factor of about 2.7, the future cumulative power generation time T_{aj} ($j=1$ to n) of the fuel cell apparatus **15A** at which the end-of-life threshold time T_E is reached.

Example 2

[0071] A life prediction method according to Example 2 of the embodiment is substantially the same as the life prediction method according to the embodiment, except for that the cumulative power generation time at which each of the power generation units 15_{ij} ($i=a$ to e , $j=1$ to n) is determined to reach the end of life is calculated based on the prediction of a decrease in voltage due to degradation of each of the power generation units 15_{ij} ($i=a$ to e , $j=1$ to n) of the fuel cell apparatus **15**.

[0072] FIG. 5 is a flowchart according to Example 2 of the embodiment, illustrating an example of an operation (method for predicting the life of each of power generation units of the fuel cell apparatus) of the life prediction apparatus in the power generation system. The operation described below may be performed by, for example, the processing section of the controller **23** reading the control program from the storage section of the controller **23**. However, the operation described below does not necessarily need to be performed by the controller **23**. An operator may perform part of the operation. The example described below is directed to a case where the operation is controlled by the controller **23**.

[0073] First, at step **S21**, a voltage V is derived from known data. The voltage V is the voltage during a time slot $[t_a, t_b]$ from time instant t_a to time instant t_b , which is a slot of time when a predetermined time has elapsed from the start of power generation by each of the power generation units 15_{ij} ($i=a$ to e , $j=1$ to n). As illustrated in FIG. 6, the time slot $[t_a, t_b]$ corresponds to a suitable voltage stabilization timing when the voltage stabilizes after the elapse of a predetermined time

from the start of power generation at rated operation. Accordingly, the voltage V may be the mean voltage during the time slot $[t_a, t_b]$. In one non-limiting example, the time from time instant t_a to time instant t_b may be about 30 minutes. The “known data” refers to the voltage at a voltage stabilization timing following the start of power generation at a suitable time between when the power generation time of each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n) is “zero”, and the current cumulative power generation time. The known data is stored in the storage section of the power generation system **10**.

[0074] Subsequently, at step **S22**, an approximation line L is derived. The approximation line L represents the correlation between the cumulative power generation time T of each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n), and the voltage V . Specifically, a power generation unit is generally known to decrease in voltage due to degradation of the power generation unit as the cumulative power generation time T of the power generation unit increases. Accordingly, the approximation line L can be obtained based on the values of the voltage V (known data) represented by circles in FIG. 7.

[0075] Subsequently, at step **S23**, the cumulative power generation time at which each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n) is determined to reach the end of life (hereinafter, “end-of-life threshold time T_E ”) is derived from the approximation line L derived at step **S22**, and from an end-of-life threshold voltage V_1 of the power generation unit. Specifically, since the end-of-life threshold voltage V_1 is a value determined in advance for the power generation system **10**, the end-of-life threshold time T_E can be found from the approximation line L and the end-of-life threshold voltage V_1 as illustrated in FIG. 7.

[0076] Subsequently, at step **S24**, the end-of-life time of each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n) is predicted from the end-of-life threshold time T_E derived at step **S23**, and from the ratio R_1 obtained at step **S1** in FIG. 3.

[0077] For each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n), the operations at steps **S21** to **S24** may be repeated every time the cumulative power generation time T increases by a predetermined amount. Information indicating the predicted end-of-life time of each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n) may be communicated to the display **40** (see FIG. 2). The information may be displayed as, for example, a year/month/day on a calendar on the display **40**.

[0078] As described above, with the life prediction method according to Example 2, for example, it is predicted, from the progression of the voltage V (actual measurement data) with respect to the cumulative power generation time of each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n), how the value of the voltage V will decrease with the passage of the cumulative power generation time in the future. Then, the cumulative power generation time at which the predicted value of the voltage V reaches the predetermined end-of-life threshold voltage V_1 can be calculated appropriately as the cumulative power generation time (the end-of-life threshold time T_E) at which each of the power generation units $15ij$ ($i=a$ to e , $j=1$ to n) is determined to reach the end of life.

[0079] The life prediction method, the life prediction apparatus **20**, and the power generation system **10** according to Example 2 may be substantially the same as those according to the embodiment or Example 1 of the embodiment, except for the characteristic features mentioned above.

Example 3

[0080] A life prediction method according to Example 3 of the embodiment is substantially the same as the life prediction method according to the embodiment, except for that the predetermined period includes time slots TS , and that the ratio R_1 is calculated by dividing a first value by a second value, the first value being the summation, over the predetermined period, of the number of power generation units that generate power in each of the time slots TS , the second value being the total number of power generation units multiplied by the total number of the time slots TS included in a predetermined period.

[0081] FIG. 8 illustrates an example of a power generation plan for the power generation system

according to Example 3 of the embodiment, for explaining an example of a method for determining the ratio R1 of the number of power generation units that generate power out of the total number of power generation units in a predetermined period.

[0082] FIG. 8 illustrates, for each of the first day, a portion of the second day, and a portion of the seventh day for a case where the predetermined period in the power generation plan stored in the storage section of the controller 23 is one week, a subset of power generation units determined as being required to generate power (hatched areas in FIG. 8), and a subset of power generation units determined as not being required to generate power (blank areas in FIG. 8).

[0083] As illustrated in FIG. 8, the predetermined period in the power generation plan includes the time slots TS. In one example, the duration of each time slot TS may be, but is not limited to, about several tens of minutes to several hours.

[0084] The number-based operating ratio (the ratio R1) of the fuel cell apparatus 15 corresponds to the ratio of the sum total of the number of power generation units determined as being “required to generate power” in each of the time slots TS in the predetermined period in the power generation plan, to the sum total of the number of power generation units that are “required to generate power” in each of the time slots TS assuming that all of the power generation units of the fuel cell apparatus 15 are “required to generate power” in all of the time slots in the predetermined period in the power generation plan.

[0085] That is, with the life prediction method according to Example 3, the number-based operating ratio (ratio R1) of the fuel cell apparatus 15 can be calculated from the following values: the summation of the number of power generation units that are “required to generate power” in each of the time slots TS; and the summation of the total number of power generation units in each of the time slots TS.

[0086] Now, assuming that all of power generation units determined as being “required to generate power” generate power at rated operation over the predetermined period in the power generation plan, the number-based operating ratio (ratio R1) of the fuel cell apparatus 15 can be also calculated as described below based on the amount of power generation according to the power generation plan.

[0087] That is, the number-based operating ratio (ratio R1) of the fuel cell apparatus 15 in this case corresponds to the ratio of the total amount of power generated by power generation units determined as being “required to generate power”, to the total amount of power generated assuming that all of the power generation units of the fuel cell apparatus 15 are “required to generate power” in all of the time slots in the predetermined period in the power generation plan.

[0088] The exemplary configuration mentioned above is intended to be illustrative and not limiting. For example, the foregoing description is directed to the case where the number-based operating ratio (the ratio R1) of the fuel cell apparatus 15 is calculated by using the number of power generation units that are determined as being “required to generate power” in the power generation plan stored in the storage section of the controller 23. Alternatively, for example, the ratio R1 may be calculated by using the number of power generation units that have actually generated power. Further, the predetermined period in the power generation plan is not limited to one week.

[0089] The life prediction method, the life prediction apparatus 20, and the power generation system 10 according to Example 3 may be substantially the same as those according to any one of the embodiment and Examples 1 and 2 of the embodiment, except for the characteristic features mentioned above.

[0090] The embodiment, and Examples 1 to 3 of the embodiment may be combined with each other as long as no conflict occurs therebetween. From the foregoing description, numerous improvements and other embodiments of the present disclosure would be apparent to persons skilled in the art. Therefore, the foregoing description should be construed as illustrative only, and has been provided for the purpose of teaching persons skilled in the art the best modes for carrying out the present disclosure. Details of the structure and/or function according to the present

disclosure can be changed substantially without departing from the spirit of the present disclosure. [0091] An aspect of the present disclosure is applicable to a life prediction method, a life prediction apparatus, and a power generation system that make it possible to predict, with higher accuracy than in the related art, the time when each of power generation units belonging to a power generation group and including a fuel cell reaches the end of life.

Claims

1. A life prediction method for a fuel cell apparatus including power generation units, the power generation units each including a fuel cell, the life prediction method comprising: predicting a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period.
 2. The life prediction method according to claim 1, wherein the cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life is calculated based on a prediction of a decrease in voltage due to degradation of the power generation unit of the fuel cell apparatus.
 3. The life prediction method according to claim 1, wherein the predetermined time includes time slots, and wherein the ratio is calculated as a first value divided by a second value, the first value being a summation, over the predetermined period, of a number of the power generation units that generate power in each of the time slots, the second value being the total number of the power generation units multiplied by a total number of the time slots included in the predetermined period.
 4. A life prediction apparatus for a fuel cell apparatus including power generation units, the power generation units each including a fuel cell, the life prediction apparatus comprising: a controller that predicts a time when each of the power generation units of the fuel cell apparatus reaches an end of life, from a cumulative power generation time of the power generation unit of the fuel cell apparatus at which the power generation unit is determined to reach the end of life, and from a ratio of a number of the power generation units that generate power out of a total number of the power generation units of the fuel cell apparatus in a predetermined period; and a communicator that communicates information to a display, the information indicating the predicted time when the power generation unit of the fuel cell apparatus reaches the end of life.
 5. A power generation system comprising: a fuel cell apparatus including power generation units, the power generation units each including a fuel cell; and the life prediction apparatus according to claim 4.
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